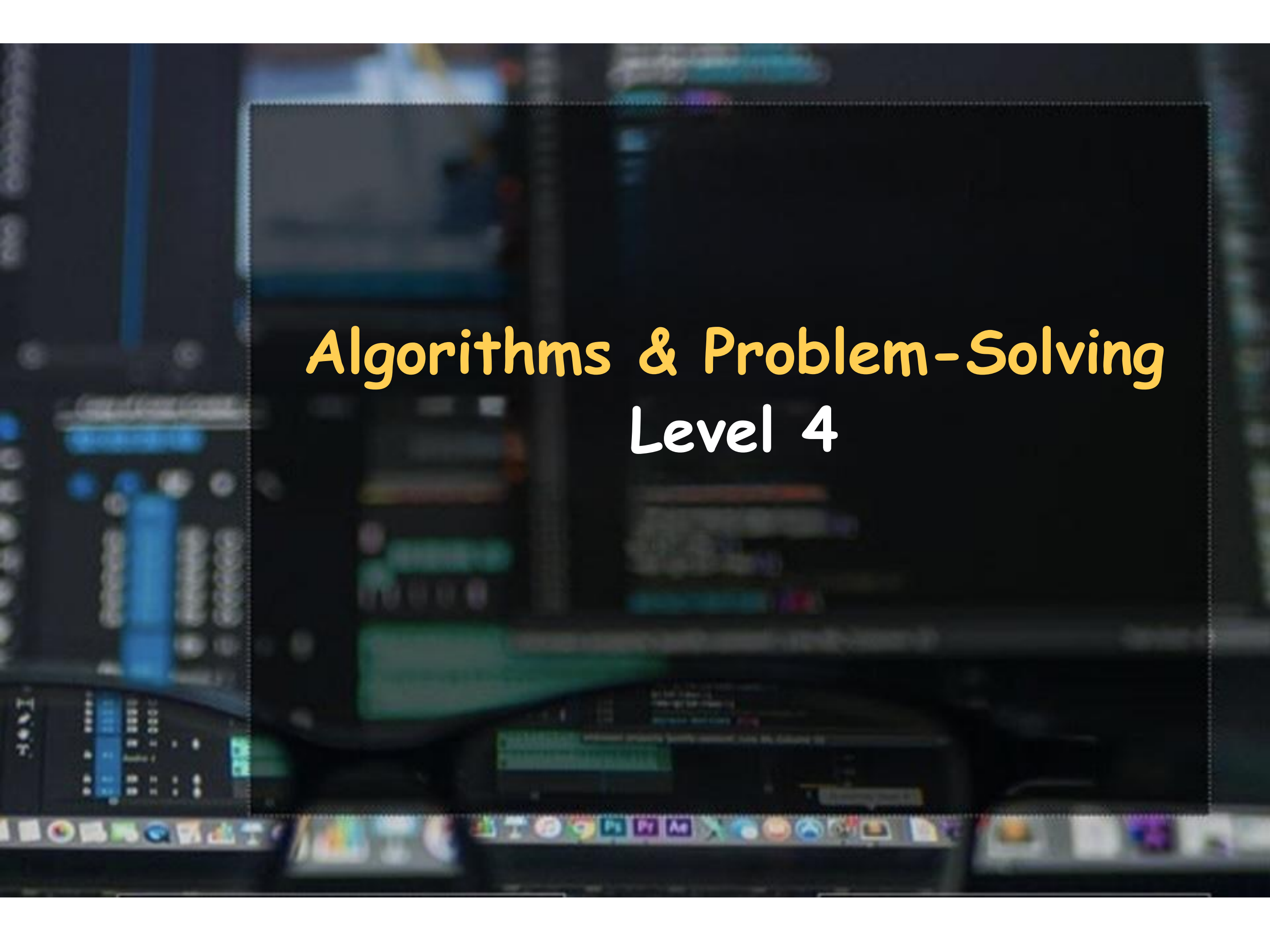
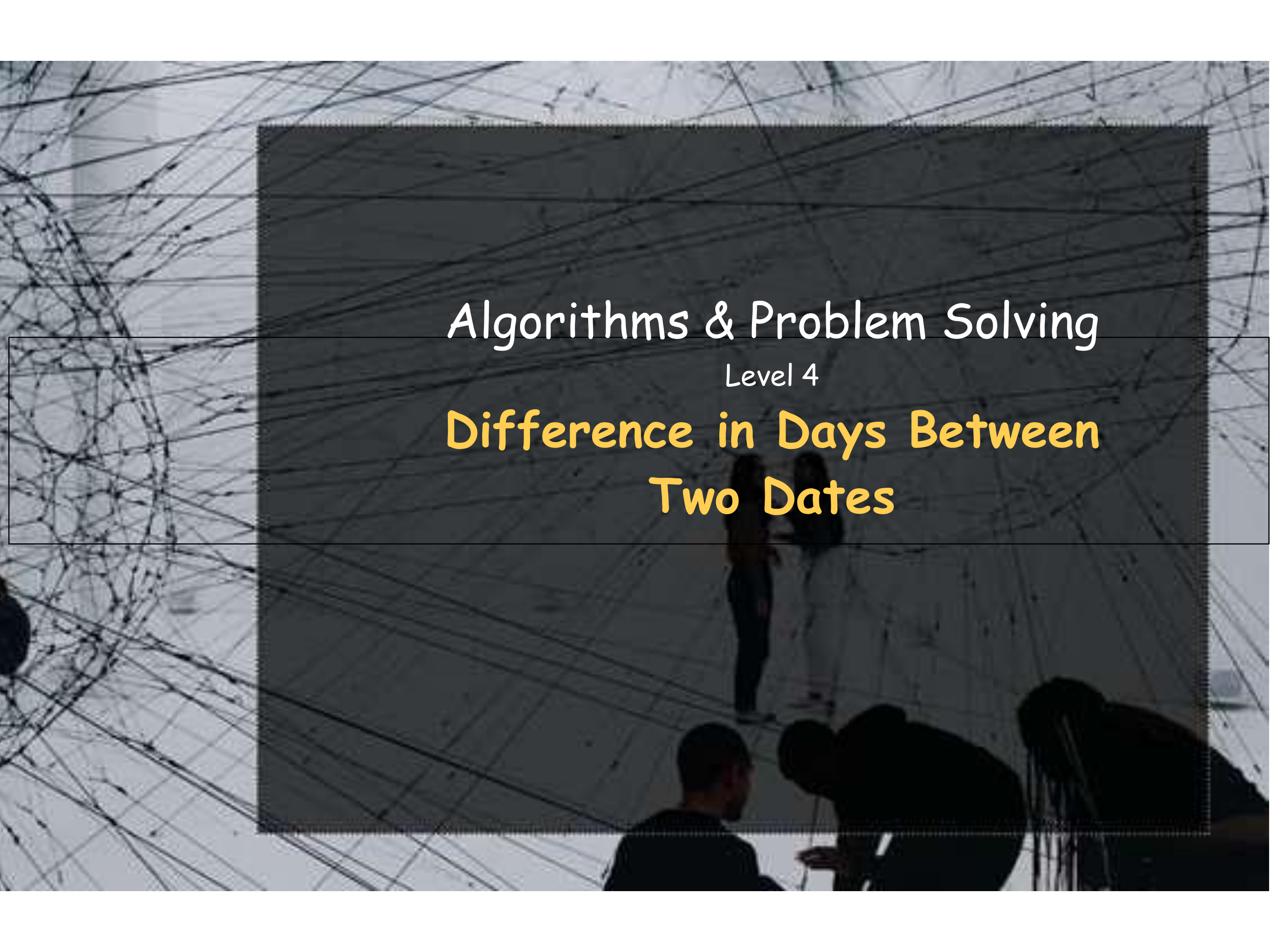




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. The overall tone is professional and technical.

Algorithms & Problem Solving

Level 4

Difference in Days Between Two Dates



Definition

- The goal is a program that reads two dates (Date1 and Date2) from the user, then calculates the difference between them in days, with the option to include or exclude the last day.



Explanation

- The program aims to accurately determine the number of days between two specific dates, including transitions between months and years, while also taking leap years into account.

Procedures

1 Defining the sDate structure:

- Contains:

 - Year — Year

 - Month — Month

 - Day — Day

2 Functions to read user input:

- ReadDay()

- ReadMonth()

- ReadYear()

- ReadFullDate() — Combines all three into a variable of type sDate

3 Leap year check:

- IsLeapYear()

- Returns true if the year is divisible by 4 but not by 100, or divisible by 400

4 Comparing two dates:

- IsDate1BeforeDate2()

- Returns true if Date1 is earlier than Date2 (i.e., earlier in time)

5 Determining the number of days in a month:

- NumberOfDaysInAMonth()

- Determines the number of days according to the month and year, with the correct handling of February in a leap year.

⑥ Incrementing the date by one day:

- `IncreaseDateByOneDay()`i
- If the last day is in the last month of the year → rolls to January 1st of the following year
- If it's only the last day of the month → rolls to the first day of the following month
- Otherwise → increments the day only

⑦ Calculating the difference in days between two dates:

- `GetDifferenceInDays(date1, date2, includeEndDay)`i
- Uses a while loop to increment date1 by one day at a time until it reaches date2
- Each increment represents one day, which is added to the days counter
- When `includeEndDay = true`, an additional day is added to the result



Example:

Please enter a Day? 8

Please enter a Month? 6

Please enter a Year? 1980

Please enter a Day? 20

Please enter a Month? 5

Please enter a Year? 1981

Difference is: 346 Day(s).

Difference (Including End Day) is: 347 Day(s).



Note:

- The first date (Date1) must be earlier than the second (Date2), otherwise the program will return zero.
- The program is based on daily repetition, so it may be slightly slower when calculating very long time periods, but it is more accurate and simpler.

Explanation:

- The program simulates the actual passage of days from one date to another, instead of direct mathematical calculation, ensuring accuracy even when switching between leap months and years.

Warning:

-  If an incorrect date is entered (such as month 13 or day 32), the program will not verify the input and may give an illogical result.

Summary

- ✓ The program calculates the difference between two dates with high accuracy.
- ✓ It takes into account leap years and month changes.

Thank you 😊